Yost 2019 BCC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=5) integrates all 5 patients: same-sample neighbour fraction 0.94→0.41, iLISI median 1.02→3.21 of 5. One connected component (0 islands). Mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

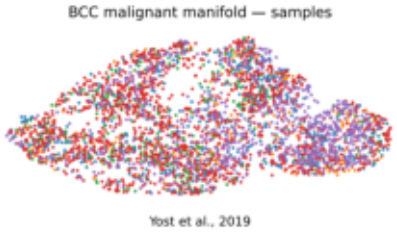
Single (Gast/pSom)→Crest developmental gradient. Gast & pSom co-vary (smoothed r=+0.61), both oppose Crest (pSom↔Crest r=−0.84, Gast↔Crest r=−0.82). Along pseudotime Gast/pSom fall (ρ −0.43/−0.75), Crest rises to a terminal tip (decile pt=0.84); joint pSom&Crest enrichment 0.32 (<1 = inverse gradient, not a hub).

Gate 3 — cross-repression → NA

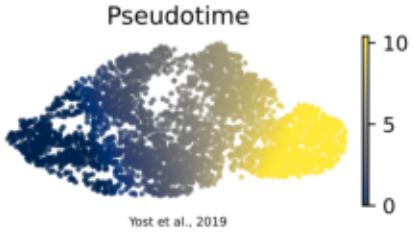
Not applicable — the manifold is non-branching (principal curve, 2 tips / 0 forks). No bifurcation, so the branching-only hub cross-repression test is not run.

STEP FIGURES

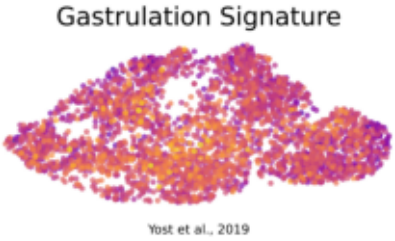
Manifold — samples



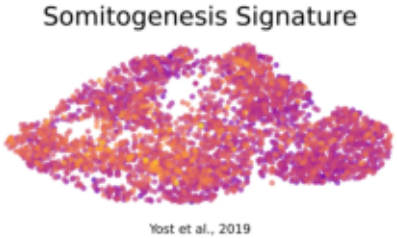
Pseudotime



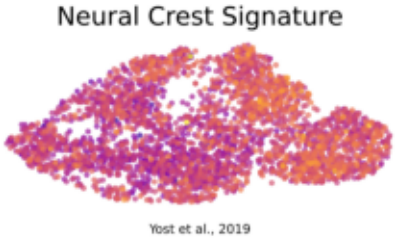
Gastrulation



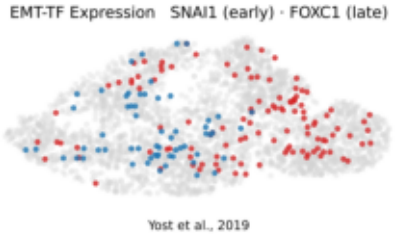
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Cutaneous basal cell carcinoma (Yost et al., 2019), 10x — 3,391 malignant cells over 5 patients (5/16 kept at ≥110-cell floor); single low-cycling body (~11% cycling). Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation/somitogenesis → neural-crest axis, Crest terminal. G1 GO / G2 GO / G3 NA. EMT-TFs SNAI1 (early/root) → FOXC1 (late/terminus).

Parameter choices

Sample threshold	110 cells (geo-mean of cliff 254→48, ratio 5.29) → 5/16, 3,391 cells	**Overdispersed genes**	1007 (scFates find_overdispersed)
Integration	Harmony on sample θ=5 (mixing plateau; patient-specific batch)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 0 islands removed
Geometry	non-branching curve N=20, λ=0.3, μ=100; 2 tips / 0 forks (2.92)	**Root**	tip 2 = Gast/pSom-high, Crest-low origin (from signature order)
Module means (raw)	Gast −0.015 · pSom +0.046 · Crest −0.011 (Linear-pSom/Crest)	**EMT-TFs**	SNAI1 (early/root) → FOXC1 (late/terminus); detection-gated; FOXC/PRRX in panel